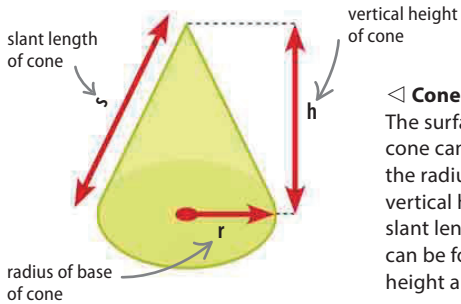


Surface and volume area

The illustrations below show three-dimensional shapes and the formulas for calculating their surface areas and their volumes. In the formulas, two letters together means that they are multiplied together, for example "2r" means "2" multiplied by "r". Pi (π) is 3.14 (to 2 decimal places).

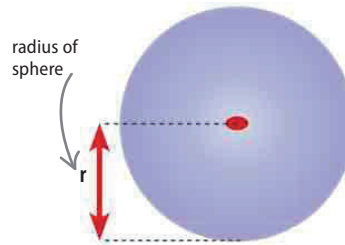


◁ Cone

The surface area of a cone can be found from the radius of its base, its vertical height, and its slant length. The volume can be found from the height and radius.

$$\text{surface area} = \pi r s + \pi r^2$$

$$\text{volume} = \frac{1}{3} \pi r^2 h$$

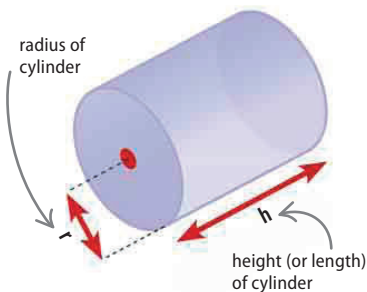


◁ Sphere

The surface area and volume of a sphere can be found when only its radius is known, because pi is a constant number (equal to 3.14, to 2 decimal places).

$$\text{surface area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3} \pi r^3$$

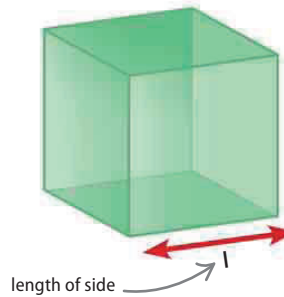


◁ Cylinder

The surface area and volume of a cylinder can be found from its radius and height (or length).

$$\text{surface area} = 2\pi r^2 + 2\pi r h$$

$$\text{volume} = \pi r^2 h$$

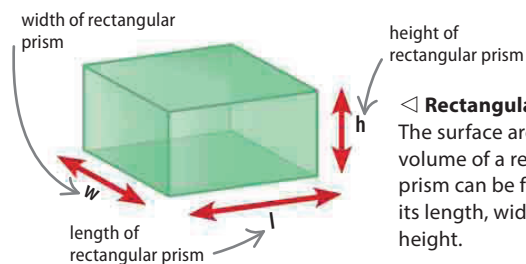


◁ Cube

The surface area and volume of a cube can be found when only the length of its sides is known.

$$\text{surface area} = 6l^2$$

$$\text{volume} = l^3$$

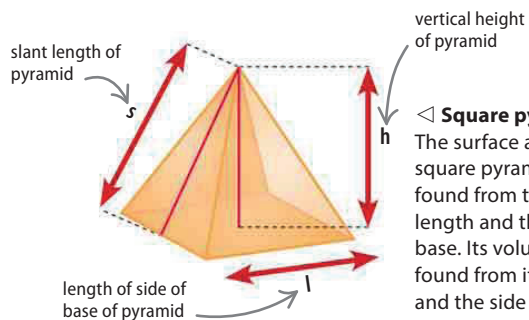


◁ Rectangular prism

The surface area and volume of a rectangular prism can be found from its length, width, and height.

$$\text{surface area} = 2(lh + lw + hw)$$

$$\text{volume} = lwh$$



◁ Square pyramid

The surface area of a square pyramid can be found from the slant length and the side of its base. Its volume can be found from its height and the side of its base.

$$\text{surface area} = 2ls + l^2$$

$$\text{volume} = \frac{1}{3} l^2 h$$